CO250 Assignment 4

October 5–19

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[1] The following LP is in SEF:

$$\begin{aligned} \max \quad & (1,-2,0,1,3)x \\ \text{subject to} \quad & \begin{pmatrix} 1 & -1 & 2 & -1 & 0 \\ 2 & 0 & 1 & -1 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned} \tag{1}$$

Find an equivalent LP in canonical form for:

[a] the basis $\{1, 4\}$,

Answer:

$$\begin{aligned} \max \quad & (1,-2,0,1,3)x \\ \text{subject to} \quad & \begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 0 & 2 & -3 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

[b] the basis $\{3, 5\}$.

Answer:

$$\begin{aligned} \max \quad & (1,-2,0,1,3)x \\ \text{subject to} \quad & \begin{pmatrix} 1/2 & -1/2 & 1 & -1/2 & 0 \\ 3/2 & 1/2 & 0 & -1/2 & 1 \end{pmatrix} x = \begin{pmatrix} 1/2 \\ -3/2 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

In each case, state whether the basis (i.e. the corresponding basic feasible solution) is feasible.

[2] Consider the following parameterized linear program in standard equality form, where $\alpha \in \mathbb{R}$:

$$\begin{aligned} \max \quad & x_4 \\ \text{s.t.} \quad & -x_1 + x_3 = 1, \\ & x_2 + \alpha x_3 + x_4 = 0, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned} \tag{P_α}$$

[a] Find a certificate of infeasibility of (P_α) for $\alpha = 1$.

Answer:

[b] Find a certificate of unboundedness of (P_α) for $\alpha = -1$.

Answer:

[c] Determine all values of α for which (P_α) is infeasible. Provide a certificate of infeasibility for each such α.

Answer:

[d] Determine all values of α for which (P_α) is unbounded. Provide a certificate of unboundedness for each such α.

Answer:

[e] For any remaining feasible values of α, find an optimal solution $\bar{x}$ and a certificate for the optimality of $\bar{x}$.

Answer:

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[3] In this exercise, you will work through a guided application of the simplex method to an LP. You do not need to have learned all the details of the algorithm in order to complete the exercise.

Consider the following LP

$$\begin{aligned} \max \quad & (1 \quad 1 \quad 0 \quad 0)x \\ \text{s.t.} \quad & \begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ & x \geq \mathbf{0}. \end{aligned}$$

[a] Let $\bar{x}$ be the basic feasible solution with basis $\{3, 4\}$. What is $\bar{x}$?

Answer:

[b] Explain why $\bar{x}$ is not an optimal solution.

Answer:

[c] Let $x^{(\alpha)}$ be the solution to

$$\begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

such that $x_1^{(\alpha)} = \alpha$ and $x_2^{(\alpha)} = 0$. Determine the set

$$\mathcal{A} = \{\alpha : x^{(\alpha)} \text{ is a feasible solution}\}.$$

Answer:

[d] Let α^* be the maximum value of α in $\mathcal{A}$ (from part (c) you should see that the maximum exists). Show that $x^{(\alpha^*)}$ is a basic feasible solution whose objective value is greater than that for $\bar{x}$. What is the basis for $x^{(\alpha^*)}$?

Answer:

[e] Redo part (c) but swapping the role of x_1 and x_2. Let $x^{(\alpha)}$ be the solution to

$$\begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

such that $x_1^{(\alpha)} = 0$ and $x_2^{(\alpha)} = \alpha$. Determine the set

$$\mathcal{B} = \{\alpha : x^{(\alpha)} \text{ is a feasible solution}\}.$$

What does $\mathcal{B}$ imply about the three possible outcomes of this LP?

Answer:

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[4] Consider the following linear program in standard equality form:

$$\begin{aligned} \min \quad & x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 3, \\ & x_1 + 2x_2 + 3x_3 + x_4 = 6, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned} \tag{P}$$

In matrix form, the constraints are $Ax = b$, $x \geq 0$, where

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 3 \\ 6 \end{pmatrix}.$$

Let $\bar{x}$ be a feasible solution given by $(1\ 1\ 1\ 0)^T$. Let $J = \{1, 2, 3\}$ be the set of indices corresponding to the positive coordinates of $\bar{x}$.

[a] Find a nonzero vector $d \in \mathbb{R}^4$ such that

$$Ad = 0$$

and d is supported on J. That is, $d_j = 0$ for every $j \notin J$.

Answer:

[b] For $\alpha \in \mathbb{R}$, define

$$x^{(\alpha)} = \bar{x} + \alpha d.$$

Show that $Ax^{(\alpha)} = b$ for every $\alpha \in \mathbb{R}$. Determine $\mathcal{A}$, the set of all values of α for which $x^{(\alpha)}$ is feasible.

Answer:

[c] Choose α^* to be either $\max\{\alpha : \alpha \in \mathcal{A}\}$ or $\min\{\alpha : \alpha \in \mathcal{A}\}$, whichever exists. Compute the resulting feasible solution $x^* = x^{(\alpha^*)}$ and compare the number of its positive coordinates with that of $\bar{x}$.

Answer:

[d] Prove that every feasible LP in SEF has a basic feasible solution by generalising the above arguments as follows. Consider LP: $\max\{c^T x : Ax = b, x \geq \mathbf{0}\}$. Let $\bar{x}$ be a feasible solution with minimum number of positive entries. Let J be the set of indices corresponding to the positive coordinates of $\bar{x}$. Following the guidance below to show that $\{A_j : j \in J\}$ is linearly independent.

(i) Suppose on the contrary that $\{A_j : j \in J\}$ is not linearly independent. Show that there exists a nonzero d supported on J such that $Ad = 0$.

Answer:

(ii) Explain why without loss of generality we may assume that $d_j < 0$ for some $j \in J$.

Answer:

(iii) For $\alpha \in \mathbb{R}$, define

$$x^{(\alpha)} = \bar{x} + \alpha d.$$

Show that $Ax^{(\alpha)} = b$ for every $\alpha \in \mathbb{R}$.

Answer:

(iv) Let $\mathcal{A}$ be the set of all values of α for which $x^{(\alpha)}$ is feasible. Show that $\mathcal{A}$ has a maximum.

Answer:

(v) Let α^* be the maximum of $\mathcal{A}$ and let $x^* = x^{(\alpha^*)}$. Show that the number of positive coordinates in x^* is smaller than $|J|$.

Answer:

(vi) Conclude that $\{A_j : j \in J\}$ is linear independent. Explain why $\bar{x}$ is a basic feasible solution.

Answer: